

FELIPE DUENAS

duenasfd@gmail.com | (510) 828-7750 | Los Angeles, CA | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of California, Los Angeles (UCLA)

Expected March 2027

B.S. Statistics & Data Science | Minor in Data Engineering

GPA: 3.86

- Relevant Coursework: Software Engineering, Database Systems, Machine Learning, Data Mining, Generative Data Science, Cloud Computing, Neural Networks, Data Structures & Algorithms

EXPERIENCE

Amazon Web Services (AWS)

Seattle, WA

Software Engineer Intern, Data Platform

May 2026 – Present

- Engineered Airflow-orchestrated ETL pipelines transforming **3,000+ Tableau workbooks** and disparate supply-chain sources into an AI-ready knowledge base, expanding queryable data by **45%** for an internal agentic-AI platform
- Developed the platform's improved data model and built its retrieval-evaluation framework, increasing agent **accuracy by 13%** and scaling self-serve analytics to **4,000+ members** of AWS's infrastructure organization
- Architected the agent's dashboard-generation tooling using LangGraph and MCP, enabling creation of context-specific dashboards from natural-language prompts and cutting average dashboard development time from **3 days to 4 minutes**
- Analyzed usage patterns on legacy dashboards using Redshift and SQL, partnering with planning, infra, and BI stakeholders to identify **15 high-value features** guiding the platform's continued tool development

UCLA Trustworthy AI Labs

Los Angeles, CA

Backend & AI Engineer Intern – Agentic Data Science

January 2026 – May 2026

- Built an agentic data science application that autonomously plans and executes multi-step analysis pipelines from natural language prompts, achieving a **71% task completion** on the BIRD benchmark, outperforming GPT-4o by 8 percentage points
- Architected a serverless backend using Dockerized AWS Lambda behind API Gateway, provisioned via AWS CDK, to handle concurrent LLM inference requests across the agent's multi-step tool-calling pipeline
- Implemented an event-sourced persistence layer using Supabase (PostgreSQL) and S3 to solve distributed state management across the serverless multi-agent architecture, maintaining consistent agent state with **zero data loss** across **400+ test sessions**
- Created a data-centric evaluation harness spanning 34 table schemas and 120+ question types across 20+ specialized tool-calling functions, driving 3 architectural changes that improved task **accuracy by 11%** over baseline

Amazon Web Services (AWS)

Arlington, VA

Business Intelligence Engineer Intern

June 2025 – September 2025

- Developed an ensemble forecasting model to predict data center power utilization, achieving **1.9% MAE** vs. a 6.3% baseline and identifying under usage that informed a **\$400K** cost saving initiative across 12 AZs
- Engineered ELT pipelines using dbt and Redshift to process **500M+ daily telemetry events** from data center hardware, creating BI-ready data structures that helped reduce reporting time from **6 hours to 30 minutes** weekly for planners
- Designed a statistical anomaly detection system to flag capacity deviations in global S3 Glacier demand, replacing manual Excel-based monitoring workflows and catching **14 anomalies monthly** before they impacted capacity
- Built a centralized Tableau dashboard with a GenAI natural-language query interface for capacity planning, unifying outputs from 7 legacy sources and cutting data-gathering time by **4 hours** per planning project

Acer

San Jose, CA

Data Analyst Intern

June 2024 – September 2024

- Audited and standardized 20+ years of legacy data across 3 siloed systems (Oracle DB, SharePoint, Excel), building a unified data model that reduced cross-team **query time by 35%**
- Identified a **\$22K** licensing discrepancy through data reconciliation of 12,000+ IT assets across procurement records, driving resolution within one billing cycle
- Automated 3 manual monthly SLA compliance reports into Tableau dashboards for General Counsel, reducing report preparation time from **2 hours to 20 minutes** and eliminating version-control issues

SKILLS

- **Languages:** Python, SQL, Java, R, Bash
- **Pipeline & DevOps:** Spark, Airflow, dbt, Kafka, Docker, CI/CD, Git, Terraform, CDK, Data Modeling
- **Cloud:** AWS, GCP, Snowflake, Redshift
- **Generative AI:** Claude, LLM-as-a-judge, RAG, LangGraph, Prompt Engineering, MCP, Vector Databases, AI Agents
- **Machine Learning:** Regression, Classification, Clustering, NLP, Neural Networks, LLM Orchestration, PyTorch, scikit-learn